

Homework 5

Problem 1

(a) Consider the following Lagrange equations for the semi-major axis a and eccentricity e .

$$\dot{a} = \frac{2}{n\sqrt{1-e^2}} \sin i [\cos(\omega + f) + e \cos \omega] a_s,$$

$$\dot{e} = \frac{\sqrt{1-e^2} \sin i}{na} \left[\cos \omega + \frac{e + \cos f}{1 + e \cos f} \cos(\omega + f) \right] a_s.$$

Integrate them over time keeping the orbital elements on the right-hand side constant, expressing the answer in terms of the eccentric anomaly; i.e., find

$$\Delta a = \int_0^t \dot{a} d\tau = \frac{1}{n} \int_0^E \dot{a} (1 - e \cos E) dE.$$

You should find the following results:

$$\Delta a = \frac{2 \sin i}{n^2} \left[\sqrt{1-e^2} \cos \omega \sin E - \sin \omega (1 - \cos E) \right] a_s,$$

$$\Delta e = \frac{\sqrt{1-e^2} \sin i}{n^2 a} \left[\frac{3}{2} \cos \omega E - 2e \cos \omega \sin E + \frac{1}{4} \cos \omega \sin 2E - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right] a_s.$$

These represent the short-period plus secular variations expected in the orbital elements as a function of time.

$$\Delta a = \int \frac{2}{n\sqrt{1-e^2}} \sin i [\cos(\omega + f) + e \cos \omega] a_s \frac{1 - e \cos E}{n} dE$$

$$= \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \int [\cos(\omega + f) + e \cos \omega] (1 - e \cos E) dE$$

$$= \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \int [(1-e^2) \cos \omega \cos E - \sin \omega \sqrt{1-e^2} \sin E] dE$$

$$= \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \left[(1-e^2) \cos \omega \sin E - \sin \omega \sqrt{1-e^2} \cos E \right]$$

$$\Delta a = \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \left[(1-e^2) \cos \omega \sin E + \sin \omega \sqrt{1-e^2} \cos E \right] + C$$

Let $\Delta a = 0$, when $E = 0$

$$\Delta a = 0 = \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} (\sqrt{1-e^2} \sin \omega) + C \Rightarrow C = -\frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \sqrt{1-e^2} \sin \omega$$

$$\Delta a = \frac{2 \sin i a_s}{n^2 \sqrt{1-e^2}} \left[(1-e^2) \cos \omega \sin E + \sin \omega \sqrt{1-e^2} (\cos E - 1) \right]$$

$$\Delta a = \frac{2 \sin i}{n^2} \left[\sqrt{1-e^2} \cos \omega \sin E - \sin \omega (1 - \cos E) \right] a_s$$

$$\cos(\omega + f) = \cos \omega \cos f - \sin \omega \sin f$$

$$\cos f = \frac{\cos E - e}{1 - e \cos E}, \quad \sin f = \frac{\sqrt{1-e^2} \sin E}{1 - e \cos E}$$

$$\cos(\omega + f) = \cos \omega \frac{\cos E - e}{1 - e \cos E} - \sin \omega \frac{\sqrt{1-e^2} \sin E}{1 - e \cos E}$$

$$\cos(\omega + f) + e \cos \omega = \cos \omega \frac{\cos E - e}{1 - e \cos E} - \sin \omega \frac{\sqrt{1-e^2} \sin E}{1 - e \cos E} + e \cos \omega$$

$$\cos(\omega + f) + e \cos \omega = \frac{\cos \omega (\cos E - e) - \sin \omega \sqrt{1-e^2} \sin E + e \cos \omega (1 - e \cos E)}{1 - e \cos E}$$

$$= \frac{\cos \omega \cos E - \sin \omega \sqrt{1-e^2} \sin E - e^2 \cos \omega \cos E}{1 - e \cos E}$$

$$= \frac{(1-e^2) \cos \omega \cos E - \sin \omega \sqrt{1-e^2} \sin E}{1 - e \cos E}$$

$$\dot{e} = \frac{\sqrt{1-e^2} \sin i}{na} \left[\cos \omega + \frac{e + \cos f}{1 + e \cos f} \cos(\omega + f) \right] a_s \quad \text{Let } \beta = \sqrt{1-e^2} \quad dt = \frac{1 - e \cos E}{n} dE$$

$$\Rightarrow \Delta e = \frac{\beta \sin i a_s}{n a} \int \left[\cos \omega + \frac{e + \cos f}{1 + e \cos f} \cos(\omega + f) \right] (1 - e \cos E) dE$$

$$= \frac{\beta \sin i a_s}{n a} \int [\cos \omega + \cos E \cos(\omega + f)] (1 - e \cos E) dE$$

$$= \frac{\beta \sin i a_s}{n a} \int [\cos \omega - e \cos E \cos \omega + \cos E \cos(\omega + f) (1 - e \cos E)] dE$$

$$= \frac{\beta \sin i a_s}{n a} \int [\cos \omega - e \cos E \cos \omega + (1-e^2) \cos \omega \cos^2 E - \beta \sin \omega \sin E \cos E] dE$$

$$= \frac{\beta \sin i a_s}{n a} \int [\cos \omega (1 - e \cos E + (1-e^2) \cos^2 E) - \beta \sin \omega \sin E \cos E] dE$$

$$\int 1 dE = E \quad \int -e \cos E dE = -e \sin E$$

$$\int (1-e^2) \cos^2 E dE = \int (1-e^2) \left(\frac{1 + \cos 2E}{2} \right) dE$$

$$= (1-e^2) \left(\frac{E}{2} + \frac{\sin 2E}{4} \right)$$

$$\int -\beta \sin E \cos E dE \quad u = \sin E \quad du = \cos E \Rightarrow \int \sin E \cos E dE = \frac{1 - \cos 2E}{4}$$

$$\cos(\omega + f) = \cos \omega \cos f - \sin \omega \sin f$$

$$\cos f = \frac{\cos E - e}{1 - e \cos E}, \quad \sin f = \frac{\sqrt{1-e^2} \sin E}{1 - e \cos E}$$

$$e + \cos f = e + \frac{\cos E - e}{1 - e \cos E} = \frac{e(1 - e \cos E) + \cos E - e}{1 - e \cos E}$$

$$= \frac{(1-e^2) \cos E}{1 - e \cos E}$$

$$1 + e \cos f = 1 + e \frac{\cos E - e}{1 - e \cos E} = \frac{1 - e \cos E + e \cos E - e^2}{1 - e \cos E}$$

$$= \frac{(1-e^2)}{1 - e \cos E}$$

$$\Rightarrow \frac{(1-e^2) \cos E}{1 - e \cos E} \div \frac{(1-e^2)}{1 - e \cos E} = \cos E$$

$$\Rightarrow \cos(\omega + f) = \frac{(1-e^2) \cos \omega \cos E - \beta \sin \omega \sin E}{1 - e \cos E}$$

$$\cos E \cos(\omega + f) (1 - e \cos E) = (1-e^2) \cos \omega \cos^2 E - \beta \sin \omega \sin E \cos E$$

$$\Rightarrow \Delta e = \frac{\beta \sin i a}{n^2 a} \left(\cos w E - e \cos w \sin E + (1-e^2) \cos w \left(\frac{E}{2} + \frac{\sin 2E}{4} \right) - \beta \sin w \left(\frac{1 - \cos 2E}{4} \right) \right) E + \frac{1-e^2}{2} E = \frac{3-e^2}{2} E$$

$$= \frac{\beta \sin i a}{n^2 a} \left(\cos w \left(\frac{3-e^2}{2} E - e \sin E + \frac{(1-e^2)}{4} \sin 2E \right) - \frac{\beta}{4} \sin w (1 - \cos 2E) \right)$$

$$= \frac{\sqrt{1-e^2} \sin i a}{n^2 a} \left(\cos w \left(\frac{3-e^2}{2} E - e \sin E + \frac{(1-e^2)}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin w (1 - \cos 2E) \right)$$

- (b) Recall that the semi-major axis has no secular change, but the eccentricity does in general. Verify that the above equations show this by comparing:

$$\frac{n}{2\pi} \Delta a(2\pi) = \frac{2}{na} \frac{\partial \bar{R}}{\partial \sigma},$$

$$\frac{n}{2\pi} \Delta e(2\pi) = \frac{1}{na^2 e} \left[(1-e^2) \frac{\partial \bar{R}}{\partial \sigma} - \sqrt{1-e^2} \frac{\partial \bar{R}}{\partial w} \right].$$

$$\Rightarrow \bar{R} = \frac{a s \sin i}{2\pi} \left(\sin w \int_0^{2\pi} r \cos f dM + \cos w \int_0^{2\pi} r \sin f dM \right)$$

Note: $r \cos f = a(\cos E - e)$
 $r \sin f = a\sqrt{1-e^2} \sin E$
 $dM = (1-e \cos E) dE$

$$\int_0^{2\pi} r \cos f dM = \int_0^{2\pi} a(\cos E - e)(1-e \cos E) dE$$

$$= a \int_0^{2\pi} (\cos E - e \cos^2 E - e + e^2 \cos E) dE$$

$$= a \left(\int_0^{2\pi} \cos E dE - \int_0^{2\pi} e \cos^2 E dE - \int_0^{2\pi} e dE + \int_0^{2\pi} e^2 \cos E dE \right)$$

$$= a(-e\pi - e2\pi) = -3\pi a e$$

$$\int_0^{2\pi} r \sin f dM = a\sqrt{1-e^2} \int_0^{2\pi} \sin E (1-e \cos E) dE = a\sqrt{1-e^2} \left[\int_0^{2\pi} \sin E dE - \int_0^{2\pi} e \sin E \cos E dE \right] = 0$$

$$\therefore \bar{R} = \frac{a s \sin i}{2\pi} \sin w (-3\pi a e) = -\frac{3}{2} a e s \sin i \sin w \quad \frac{n}{2\pi} \Delta a(2\pi) = \frac{2}{na} \frac{\partial \bar{R}}{\partial \sigma} \quad E=2\pi$$

$$\Delta a(2\pi) = \frac{2 s \sin i}{n^2} \left[\sqrt{1-e^2} \cos w \sin 2\pi - \sin w (1 - \cos 2\pi) \right] a s = 0 \quad \& \quad \frac{\partial \bar{R}}{\partial \sigma} = 0 \quad \text{because there is no } \sigma \text{ in the expression}$$

$$\therefore \frac{n}{2\pi} \Delta a(2\pi) = \frac{2}{na} \frac{\partial \bar{R}}{\partial \sigma} \quad \checkmark \text{ no secular change for the semi-major axis}$$

$$\frac{d\bar{R}}{dw} = \frac{d}{dw} \left(-\frac{3}{2} a e s \sin i \sin w \right) = -\frac{3}{2} a e s \sin i \cos w \quad \frac{n}{2\pi} \Delta e(2\pi) = \frac{1}{na^2 e} \left((1-e^2) \frac{d\bar{R}}{d\sigma} - \sqrt{1-e^2} \frac{d\bar{R}}{dw} \right)$$

$$\frac{n}{2\pi} \Delta e(2\pi) = \frac{\sqrt{1-e^2} \sin i a}{n^2 a} \left(\cos w \left(\frac{3-e^2}{2} E - e \sin 2\pi + \frac{(1-e^2)}{4} \sin 4\pi \right) - \frac{\sqrt{1-e^2}}{4} \sin w (1 - \cos 4\pi) \right)$$

$$= \frac{\sqrt{1-e^2} \sin i a}{n^2 a} \cos w \left(\frac{3-e^2}{2} 2\pi \right)$$

$$= \frac{\sqrt{1-e^2} \sin i a}{na} \cos w \frac{3}{2}$$

$$= \frac{3\sqrt{1-e^2} \sin i a \cos w}{2na}$$

$$\frac{1}{na^2 e} \left((1-e^2) \frac{d\bar{R}}{d\sigma} - \sqrt{1-e^2} \frac{d\bar{R}}{dw} \right) = \frac{1}{na^2 e} \left((1-e^2)(0) - \sqrt{1-e^2} \left(-\frac{3}{2} a e s \sin i \cos w \right) \right)$$

$$= \frac{1}{2na} \left(3\sqrt{1-e^2} \sin i \cos w \right)$$

$$= \frac{3\sqrt{1-e^2} \sin i \cos w}{2na}$$

$$\therefore \frac{n}{2\pi} \Delta e(2\pi) = \frac{1}{na^2 e} \left((1-e^2) \frac{d\bar{R}}{d\sigma} - \sqrt{1-e^2} \frac{d\bar{R}}{dw} \right) \quad \text{eccentricity does have a secular change in general}$$

- (c) Now compare your solution from Part (a) to a numerical integration using your previous code (Homework 4, Problem 3). To be as general as possible, start from a non-circular orbit (maybe use $e = 0.1$), non-planar (or $i \neq 0$), with a non-zero argument of periapsis. Remember to have the constant acceleration act along the z -axis for these runs. Compare the numerically found osculating orbit for all 6 orbital elements against the above theory for the semi-major axis and eccentricity. Run the motion over a few orbits and compare and contrast how well the amplitude variations in these elements compares with the numerical amplitudes. If you see changes between these two, comment on what they may be due to.

Check code & plots below

The semi-major axis shows only periodic oscillations with no secular drift matching the analytical prediction for part B. The eccentricity exhibits both periodic oscillations and a secular increase over time, also consistent with the theory. The remaining orbital elements show small long-term changes due to the constant perturbing acceleration acting in the inertial z -direction. Small discrepancies between the numerical and analytical results are expected because the analytical derivation assumes constant orbital elements and neglects higher-order perturbation effects.

- (d) Now analytically compute the average of the semi-major axis short-period term, and the average of the eccentricity short-period term using the following formula:

$$\overline{\Delta a} = \frac{1}{2\pi} \int_0^{2\pi} \Delta a(E) dM,$$

$$\overline{\Delta e_p} = \frac{1}{2\pi} \int_0^{2\pi} \left[\Delta e(E) - \frac{1}{n} \dot{e} M \right] dM.$$

$$dM = (1 - e \cos E) dE$$

Compare these mean values with the mean values you get from your numerical simulations.

$$\Delta a = \frac{2 \sin i}{n^2} \left[\sqrt{1-e^2} \cos \omega \sin E - \sin \omega (1 - \cos E) \right] a_s$$

$$\Rightarrow \overline{\Delta a} = \frac{2 \sin i}{n^2} a_s \frac{1}{2\pi} \int_0^{2\pi} (\sqrt{1-e^2} \cos \omega \sin E - \sin \omega (1 - \cos E)) (1 - \cos E) dE$$

$$\sqrt{1-e^2} \cos \omega \int_0^{2\pi} \sin E (1 - \cos E) dE - \sin \omega \left(\int_0^{2\pi} \sin E dE - \int_0^{2\pi} \cos E dE \right) = 0$$

$$\sin \omega \int_0^{2\pi} (1 - \cos E - \cos E + \cos^2 E) dE = \sin \omega (2\pi + \pi) = 3\pi \sin \omega$$

$$\Rightarrow \overline{\Delta a} = \frac{2 \sin i}{n^2} a_s \frac{1}{2\pi} \sin \omega (2\pi + \pi) = \frac{\sin i}{n^2} a_s \sin \omega (3\pi) = \frac{3\pi \sin i \sin \omega a_s}{n^2}$$

$$\Delta e = \frac{\sqrt{1-e^2} \sin i}{n^2 a} \left(\cos \omega \left(\frac{3-e^2}{2} E - e \sin E + \frac{(1-e^2)}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right)$$

$$\dot{e} = \frac{3 \sqrt{1-e^2} \sin i \cos \omega}{2 n a} a_s \Rightarrow \frac{1}{n} \dot{e} M = \frac{3 \sqrt{1-e^2} \sin i \cos \omega}{2 n a} a_s M \quad \text{Let } B = \frac{\sqrt{1-e^2} \sin i}{n^2 a}$$

$$\Delta e = B \left(\cos \omega \left(\frac{3-e^2}{2} E - e \sin E + \frac{(1-e^2)}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right)$$

$$\Rightarrow \frac{1}{n} \dot{e} M = B \left(\frac{3}{2} \cos \omega M \right), \quad M = E - e \sin E$$

$$- B \left(\frac{3}{2} \cos \omega (E - e \sin E) \right) \quad \Delta e_p = \Delta e - \frac{1}{n} \dot{e} M$$

$$\Rightarrow \Delta e_p = B \left[\cos \omega \left(\frac{3-e^2}{2} E - \cancel{2e \sin E} + \frac{1}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) - \frac{3}{2} \cos \omega (E - \cancel{e \sin E}) \right]$$

$$= B \left[\cos \omega \left(\frac{1}{2} e \sin E + \frac{1}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right]$$

$$\Rightarrow \overline{\Delta e_p} = \frac{1}{2\pi} \int_0^{2\pi} \Delta e_p dM \quad dM = (1 - e \cos E) dE$$

$$= \frac{1}{2\pi} \int_0^{2\pi} B \left[\cos \omega \left(\frac{1}{2} e \sin E + \frac{1}{4} \sin 2E \right) - \frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right] (1 - \cos E) dE$$

$$= \frac{B}{2\pi} \int_0^{2\pi} \left[-\frac{\sqrt{1-e^2}}{4} \sin \omega (1 - \cos 2E) \right] (1 - \cos E) dE$$

$$= -\frac{B}{2\pi} \frac{\sqrt{1-e^2}}{4} \sin w \int_0^{2\pi} (1-\cos 2E)(1-e\cos E) dE$$

$$\int_0^{2\pi} (1-e\cos E - \cos 2E + e\cos 2E \cos E) dE = \int_0^{2\pi} 1 - \int_0^{2\pi} e\cos E - \int_0^{2\pi} \cos 2E + \int_0^{2\pi} e\cos 2E \cos E dE \quad \cos E \cos 2E = \frac{1}{2}[\cos 3E + \cos E]$$

$$= 2\pi + \int_0^{2\pi} \frac{e}{2} [\cos 3E + \cos E] dE = 2\pi$$

$$\Rightarrow \Delta \bar{e}_p = \frac{-B}{2\pi} \frac{\sqrt{1-e^2}}{4} \sin w (2\pi) = \frac{\sqrt{1-e^2} \sin i a s \sin w \sqrt{1-e^2}}{4n^2 a} = \frac{-(1-e^2) \sin i \sin w a s}{4n^2 a}$$

HW5 Part (d) Mean Short-Period Comparison

Semi-major axis short-period mean:

Theory = -7.4246212025e-05

Numerical = -7.4187168508e-05

Difference = 5.9043516116e-08

Eccentricity short-period mean:

Theory = -8.7504464172e-06

Numerical = -4.6100227859e-06

Difference = 4.1404236313e-06

The a comparison is very close. The e comparison is not as close because the numerical eccentricity includes higher-order effects and element coupling that the first-order analytical short-period expression neglects.


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In [1]: import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp

# HW5 Part (c)
# Constant z-direction acceleration perturbation
# Compare numerical osculating elements with theory

mu = 1.0

a0 = 1.0
e0 = 0.1
i0 = np.deg2rad(30.0)
Omega0 = np.deg2rad(40.0)
omega0 = np.deg2rad(45.0)
M0 = 0.0

a_s = 1e-4
a_pert = np.array([0.0, 0.0, a_s])

n0 = np.sqrt(mu / a0**3)
T0 = 2 * np.pi / n0

num_orbits = 5
t_final = num_orbits * T0
t_eval = np.linspace(0, t_final, 3000)

# Kepler solvers
def solve_kepler_wrapped(M, e, tol=1e-12):
    M = np.mod(M, 2*np.pi)

    if np.isscalar(M):
        E = M
    else:
        E = M.copy()

    for _ in range(50):
        f = E - e*np.sin(E) - M
        fp = 1 - e*np.cos(E)
        dE = -f / fp
        E = E + dE

        if np.max(np.abs(dE)) < tol:
            break

    return E

def solve_kepler_unwrapped(M, e, tol=1e-12):
    if np.isscalar(M):
        E = M
    else:
        E = M.copy()

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for _ in range(50):
    f = E - e*np.sin(E) - M
    fp = 1 - e*np.cos(E)
    dE = -f / fp
    E = E + dE

    if np.max(np.abs(dE)) < tol:
        break

return E

# Convert Orbital Elements -> Cartesian State
def elements_to_rv(a, e, inc, Omega, omega, M, mu):
    E = solve_kepler_wrapped(M, e)

    r_pf = np.array([
        a * (np.cos(E) - e),
        a * np.sqrt(1 - e**2) * np.sin(E),
        0.0
    ])

    r_mag = a * (1 - e*np.cos(E))

    v_pf = np.array([
        -np.sqrt(mu*a) * np.sin(E) / r_mag,
        np.sqrt(mu*a) * np.sqrt(1 - e**2) * np.cos(E) / r_mag,
        0.0
    ])

    c0, s0 = np.cos(Omega), np.sin(Omega)
    ci, si = np.cos(inc), np.sin(inc)
    cw, sw = np.cos(omega), np.sin(omega)

    R3_0 = np.array([
        [c0, -s0, 0],
        [s0, c0, 0],
        [0, 0, 1]
    ])

    R1_i = np.array([
        [1, 0, 0],
        [0, ci, -si],
        [0, si, ci]
    ])

    R3_w = np.array([
        [cw, -sw, 0],
        [sw, cw, 0],
        [0, 0, 1]
    ])

    Q = R3_0 @ R1_i @ R3_w

    r_vec = Q @ r_pf
    v_vec = Q @ v_pf

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    return r_vec, v_vec

# Convert Cartesian State -> Orbital Elements
def rv_to_elements(r, v, mu):
    r_mag = np.linalg.norm(r)
    v_mag = np.linalg.norm(v)

    h = np.cross(r, v)
    h_mag = np.linalg.norm(h)

    k_hat = np.array([0.0, 0.0, 1.0])
    N = np.cross(k_hat, h)
    N_mag = np.linalg.norm(N)

    e_vec = np.cross(v, h)/mu - r/r_mag
    e = np.linalg.norm(e_vec)

    energy = v_mag**2 / 2 - mu / r_mag
    a = -mu / (2 * energy)

    inc = np.arccos(h[2] / h_mag)

    Omega = np.arctan2(N[1], N[0])
    Omega = np.mod(Omega, 2*np.pi)

    omega = np.arccos(np.dot(N, e_vec) / (N_mag * e))
    if e_vec[2] < 0:
        omega = 2*np.pi - omega

    f = np.arccos(np.dot(e_vec, r) / (e * r_mag))
    if np.dot(r, v) < 0:
        f = 2*np.pi - f

    E = 2 * np.arctan2(
        np.sqrt(1 - e) * np.sin(f/2),
        np.sqrt(1 + e) * np.cos(f/2)
    )

    M = E - e*np.sin(E)
    M = np.mod(M, 2*np.pi)

    return a, e, inc, Omega, omega, M

# Dynamics
def dynamics(t, y):
    r = y[:3]
    v = y[3:]

    r_mag = np.linalg.norm(r)

    a_grav = -mu * r / r_mag**3
    a_total = a_grav + a_pert

    return np.hstack((v, a_total))

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# Initial State
r0, v0 = elements_to_rv(
    a0, e0, i0, Omega0, omega0, M0, mu
)

y0 = np.hstack((r0, v0))

# Integrate
sol = solve_ivp(
    dynamics,
    [0, t_final],
    y0,
    t_eval=t_eval,
    rtol=1e-10,
    atol=1e-12
)

# Numerical Osculating Elements
a_num = []
e_num = []
i_num = []
Omega_num = []
omega_num = []
M_num = []

for k in range(len(sol.t)):
    r = sol.y[:3, k]
    v = sol.y[3:, k]

    a, e, inc, Omega, omega, M = rv_to_elements(r, v, mu)

    a_num.append(a)
    e_num.append(e)
    i_num.append(inc)
    Omega_num.append(Omega)
    omega_num.append(omega)
    M_num.append(M)

a_num = np.array(a_num)
e_num = np.array(e_num)
i_num = np.array(i_num)
Omega_num = np.array(Omega_num)
omega_num = np.array(omega_num)
M_num = np.array(M_num)

# Theory from Part (a)
M_theory = n0 * sol.t

# Use unwrapped eccentric anomaly so E grows continuously over many orbits.
E_theory = solve_kepler_unwrapped(M_theory, e0)

beta = np.sqrt(1 - e0**2)

Delta_a = (
    (2 * np.sin(i0) / n0**2)
    *

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```

        (
            beta * np.cos(omega0) * np.sin(E_theory)
            -
            np.sin(omega0) * (1 - np.cos(E_theory))
        )
        * a_s
    )

Delta_e = (
    (beta * np.sin(i0) / (n0**2 * a0))
    *
    (
        np.cos(omega0)
        *
        (
            (3/2) * E_theory
            -
            2 * e0 * np.sin(E_theory)
            +
            (1/4) * np.sin(2 * E_theory)
        )
        -
        (beta / 4)
        * np.sin(omega0)
        * (1 - np.cos(2 * E_theory))
    )
    * a_s
)

a_theory = a0 + Delta_a
e_theory = e0 + Delta_e

# Plot 3D Trajectory
fig = plt.figure(figsize=(8, 6))
ax = fig.add_subplot(111, projection='3d')

ax.plot(sol.y[0], sol.y[1], sol.y[2], linewidth=1.5)
ax.scatter(sol.y[0], sol.y[1], sol.y[2], marker='o', s=60, label='Start')
ax.scatter(sol.y[0], sol.y[1], sol.y[2], marker='s', s=60, label='End')

ax.set_title('3D Numerical Trajectory')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_zlabel('z')
ax.legend()

plt.tight_layout()

# Plot Orbital Elements
time_orbits = sol.t / T0

fig, axs = plt.subplots(3, 2, figsize=(14, 14))
axs = axs.flatten()

axs[0].plot(time_orbits, a_num, label='Numerical')
axs[0].plot(time_orbits, a_theory, '--', label='Theory')

```

```

axs[0].set_title('Semi-major Axis')
axs[0].set_ylabel('a')
axs[0].legend()

axs[1].plot(time_orbits, e_num, label='Numerical')
axs[1].plot(time_orbits, e_theory, '--', label='Theory')
axs[1].set_title('Eccentricity')
axs[1].set_ylabel('e')
axs[1].legend()

axs[2].plot(time_orbits, np.rad2deg(i_num))
axs[2].set_title('Inclination')
axs[2].set_ylabel('i [deg]')

axs[3].plot(time_orbits, np.rad2deg(Omega_num))
axs[3].set_title('RAAN')
axs[3].set_ylabel('RAAN [deg]')

axs[4].plot(time_orbits, np.rad2deg(omega_num))
axs[4].set_title('Argument of Periapsis')
axs[4].set_ylabel('omega [deg]')

axs[5].plot(time_orbits, np.rad2deg(M_num))
axs[5].set_title('Mean Anomaly')
axs[5].set_ylabel('M [deg]')

for ax in axs:
    ax.set_xlabel('Time [orbits]')
    ax.grid(True)

fig.subplots_adjust(
    hspace=0.75,
    wspace=0.35,
    top=0.92,
    bottom=0.07
)

plt.show()

# Print comparison
print('\nHW5 Part (c) Comparison\n')

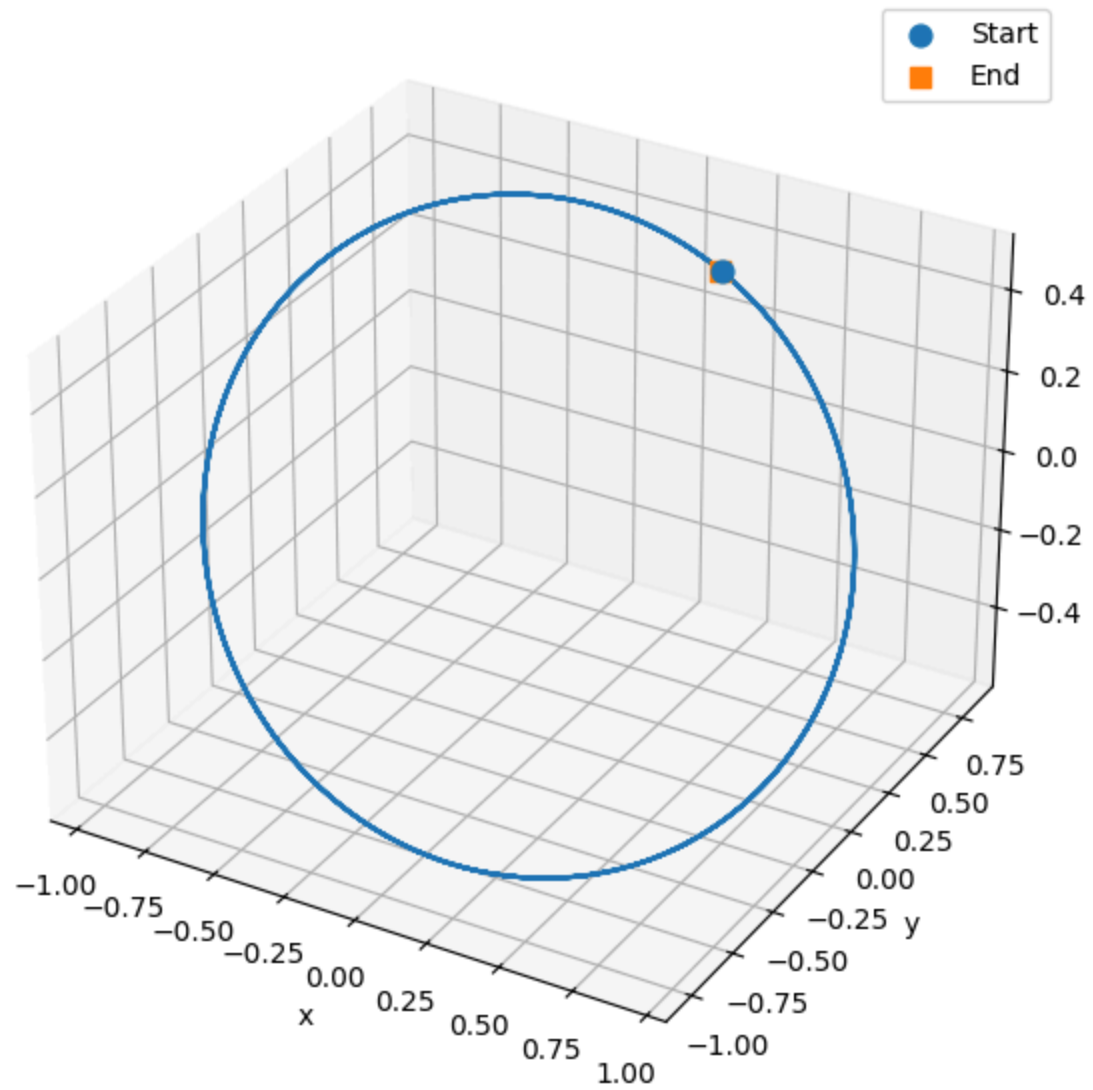
print(f'Initial semi-major axis a0 = {a0}')
print(f'Initial eccentricity e0 = {e0}')
print(f'Perturbing acceleration a_s = {a_s}\n')

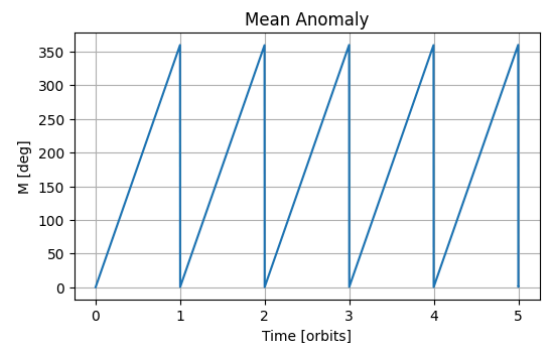
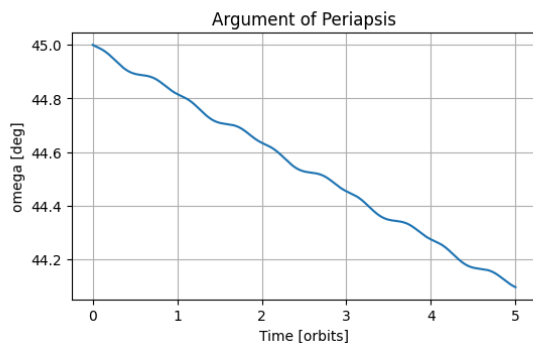
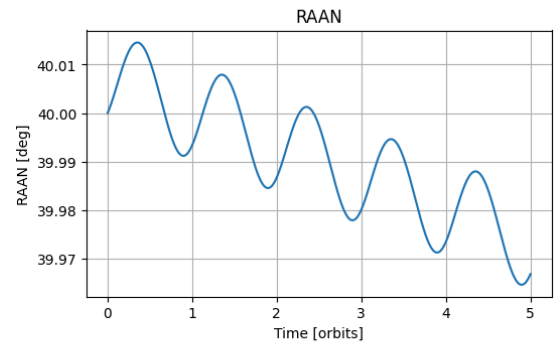
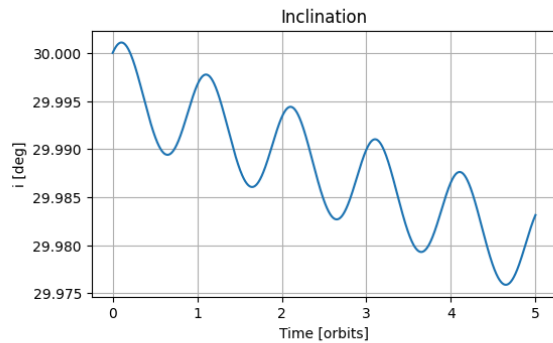
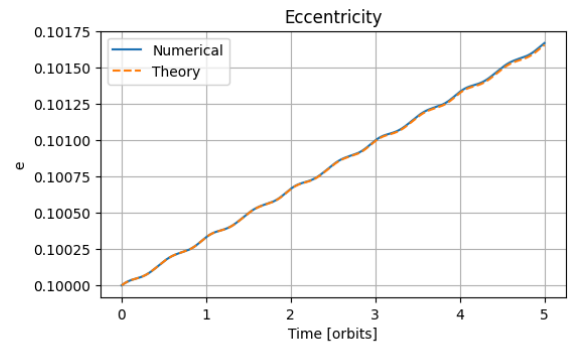
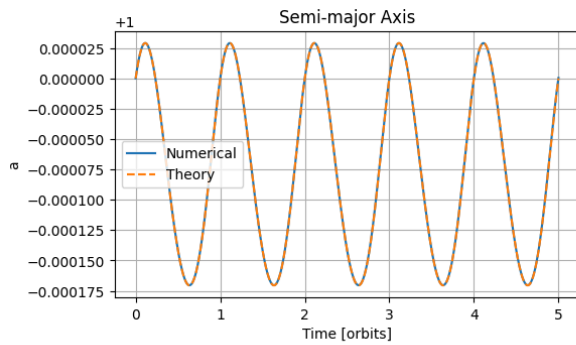
print('Semi-major axis comparison:')
print(f'Numerical min/max = {np.min(a_num):.8f}, {np.max(a_num):.8f}')
print(f'Theory min/max = {np.min(a_theory):.8f}, {np.max(a_theory):.8f}\n')

print('Eccentricity comparison:')
print(f'Numerical min/max = {np.min(e_num):.8f}, {np.max(e_num):.8f}')
print(f'Theory min/max = {np.min(e_theory):.8f}, {np.max(e_theory):.8f}')

```


3D Numerical Trajectory





HW5 Part (c) Comparison

Initial semi-major axis $a_0 = 1.0$

Initial eccentricity $e_0 = 0.1$

Perturbing acceleration $a_s = 0.0001$

Semi-major axis comparison:

Numerical min/max = 0.99982962, 1.00002904

Theory min/max = 0.99982954, 1.00002904

Eccentricity comparison:

Numerical min/max = 0.10000000, 0.10167032

Theory min/max = 0.10000000, 0.10165773

In [4]: *# HW5 Part (d): Mean short-period values*

import numpy as np

Re-define constants in case this cell is run independently

```

mu = 1.0

a0 = 1.0
e0 = 0.1
i0 = np.deg2rad(30.0)
omega0 = np.deg2rad(45.0)

a_s = 1e-4

n0 = np.sqrt(mu / a0**3)

# Analytical averages from Part (d)
Delta_a_mean_theory = -(
    (2 + e0)
    * np.sin(i0)
    * np.sin(omega0)
    / n0**2
) * a_s

Delta_ep_mean_theory = -(
    (1 - e0**2)
    * np.sin(i0)
    * np.sin(omega0)
    / (4 * n0**2 * a0)
) * a_s

# Secular eccentricity rate from Part (b)
edot_bar = (
    3
    * np.sqrt(1 - e0**2)
    * np.sin(i0)
    * np.cos(omega0)
    / (2 * n0 * a0)
) * a_s

# Numerical short-period quantities
Delta_a_num = a_num - a0

Delta_ep_num = (e_num - e0) - edot_bar * sol.t

# Numerical mean values
Delta_a_mean_num = (
    np.trapz(Delta_a_num, sol.t)
    / (sol.t[-1] - sol.t[0])
)

Delta_ep_mean_num = (
    np.trapz(Delta_ep_num, sol.t)
    / (sol.t[-1] - sol.t[0])
)

# Print comparison
print('\nHW5 Part (d) Mean Short-Period Comparison\n')

print('Semi-major axis short-period mean:')
print(f'Theory      = {Delta_a_mean_theory:.10e}')

```

```
print(f'Numerical = {Delta_a_mean_num:.10e}')
print(f'Difference = {Delta_a_mean_num - Delta_a_mean_theory:.10e}\n')

print('Eccentricity short-period mean:')
print(f'Theory = {Delta_ep_mean_theory:.10e}')
print(f'Numerical = {Delta_ep_mean_num:.10e}')
print(f'Difference = {Delta_ep_mean_num - Delta_ep_mean_theory:.10e}')
```

HW5 Part (d) Mean Short-Period Comparison

Semi-major axis short-period mean:

Theory = -7.4246212025e-05

Numerical = -7.4187168508e-05

Difference = 5.9043516116e-08

Eccentricity short-period mean:

Theory = -8.7504464172e-06

Numerical = -4.6100227859e-06

Difference = 4.1404236313e-06

In []: